Kunwoo Park

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Research Interests

Integrated photonics; Disordered photonics; Topological photonics.

Education

Integrated M.S./Ph.D., Department of Electrical and Computer Engineering — 2024.09 – present
Seoul National University Advisor: Sunkyu Yu (Expected: 2029.08)

B.S., Department of Electrical and Computer Engineering — 2018.03 – 2024.08
Seoul National University

Military Service, Republic of Korea Army — 2020.12 – 2022.06

Research Experience

- **Integrated Photonics.** Design and experimental demonstration of photonic circuits, with theoretical work toward all-optical neural networks.
- **Disordered Photonics.** Development of hypergraph models of wave scattering for scalable design of disordered photonic materials.
- **Topological Photonics.** Investigation of non-Abelian topology for robust and programmable control of light.

Publications

Preprints

1. Jungyoon Kim†, Kyuho Kim†, **Kunwoo Park**, Namkyoo Park* & Sunkyu Yu*. Task-specific programming of chaos in neural circuits.
   arXiv:2605.19465
2. **Kunwoo Park**, Ikbeom Lee, Seungmok Youn, Gitae Lee, Namkyoo Park* & Sunkyu Yu*. Hypergraph modelling of wave scattering to speed-up material design.
   arXiv:2507.15329

Journal Articles

1. Gitae Lee†, Seungmok Youn†, Ikbeom Lee†, **Kunwoo Park**, Duhwan Hwang, Seungkyun Park, Xianji Piao*, Namkyoo Park* & Sunkyu Yu*. Non-Hermitian stealthy hyperuniformity.
   Advanced Science 13, e77054 (2026)
2. Seungmok Youn, **Kunwoo Park**, Ikbeom Lee, Gitae Lee, Namkyoo Park* & Sunkyu Yu*. Phase-sensitive engineering of optical disordered materials using heterogeneous networks.
   Advanced Optical Materials 14, e03499 (2026)
3. Kyuho Kim†, **Kunwoo Park**†, Hyungchul Park, Sunkyu Yu, Namkyoo Park* & Xianji Piao*. Programmable photonic unitary circuits for light computing.
   Nanophotonics 14, 1429 (2025)

International Conferences

1. Ikbeom Lee, **Kunwoo Park**, Seungmok Youn & Sunkyu Yu*. Inverse design of microstructures of non-Hermitian scattering systems.
   SPIE Optics + Photonics 2025, 13578-59, San Diego, California, USA, 6 August 2025
2. **Kunwoo Park**, Ikbeom Lee, Seungmok Youn & Sunkyu Yu*. Modularization of light scattering for photonic computing applications.
   SPIE Optics + Photonics 2025, 13581-25, San Diego, California, USA, 5 August 2025
3. Ikbeom Lee, **Kunwoo Park** & Sunkyu Yu*. Design of wave scattering from gain and loss disordered materials.
   CLEO-PacificRim (CLEO-PR) 2024, Mo2J-4, Incheon, Republic of Korea, 5 August 2024
4. **Kunwoo Park**, Ikbeom Lee & Sunkyu Yu*. Inverse design method of hyperuniform materials using system factorization.
   CLEO-PacificRim (CLEO-PR) 2024, Mo2J-2, Incheon, Republic of Korea, 5 August 2024

Domestic Conferences

1. **Kunwoo Park**, Gitae Lee, Ikbeom Lee, Seungmok Youn, Namkyoo Park & Sunkyu Yu. Large-Scale Optical Material Design via Hypergraph Models.
   Optics and Photonics Congress 2026, W3C-V.01, Jeju, Republic of Korea, 22 July 2026.
2. Ikbeom Lee, Gitae Lee, Seungmok Youn, **Kunwoo Park**, Duhwan Hwang, Xianji Piao, Namkyoo Park & Sunkyu Yu. Designing asymmetric scattering using gain and loss media.
   Optics and Photonics Congress 2026, TP-V.23, Jeju, Republic of Korea, 21 July 2026.
3. Duhwan Hwang, Jongheon Yeo, **Kunwoo Park**, Namkyoo Park & Sunkyu Yu. Numerical Analysis of Single-Photon Dynamics in Nonreciprocal Waveguide-Resonator Coupled Systems.
   Photonics Conference 2025, T2A-II.05, Pyeongchang, Republic of Korea, 27 November 2025.

Patents

1. Nonlinear unitary optical device.
   Sunkyu Yu, Namkyoo Park, Xianji Piao & **Kunwoo Park**
   KR 10-2859166 (Application No. 10-2025-0005056; Provisional No. 10-2024-0179951)
   Registration date: 2025.09.09; Filing date: 2025.01.13; Provisional filing date: 2024.12.05
   US application: 19/048,370; Filing date: 2025.02.07
   EU application: 25156862.2; Filing date: 2025.02.10

Grants and Fellowships

Graduate Research Encouragement Grant (석사과정생연구장려금지원사업) 2025 – present
Principal Investigator

Teaching

Application of Quantum Mechanics (Graduate), Teaching Assistant Spring 2025
Seoul National University

Topics in Electro-physics (Graduate), Teaching Assistant Spring 2025
Seoul National University

References

Available upon request.

† Co-first authors. * Corresponding authors.